



2554 - AGN Feeding and Feedback in NGC 4151

Cycle: 1, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

The discovery of a number of black hole (BH) galaxy relations has shown that the growth of supermassive BHs is closely related to the evolution of galaxies. This evidence has opened a new debate in which the fundamental questions concern the interactions between the central BH and the interstellar medium within the host galaxy, and can be addressed by studying two crucial processes: feeding and feedback. Using archival JWST NIRSpec data, we aim to characterize the circumnuclear region of NGC 4151 to study, with unprecedented detail AGN feeding and feedback. NGC 4151, as one of the nearest AGN and an extremely well-studied Seyfert galaxy, provides an outstanding opportunity to obtain meaningful constraints for models incorporating these processes to regulate BH-host galaxy co-evolution. We will achieve this by: (1) tracing outflows via ionized and coronal gas emission to establish the energetics of the outflowing gas and potential for feedback into the host galaxy, (2) tracing inflows via molecular hydrogen gas emission from which the driving inflow mechanism can be identified and an inflow rate obtained via modeling, (3) using the wide range of available H₂ lines in the near-IR to establish their connection with the obscuring torus, and (4) obtaining a molecular gas-based black hole mass estimate. JWST is uniquely capable of advancing our understanding of these processes due to its sensitivity and stable point-spread-function, which facilitates the detection of diagnostic emission lines not available in current ground based data, as well as probing these processes down to unprecedented spatial scales.

OBSERVING DESCRIPTION

One of the most important issues in astrophysics today is understanding how galaxies and their SMBHs coevolve, resulting in surprisingly tight correlations such as the MBH-* relation. AGN have emerged as likely players in this coevolution, since they are by definition actively accreting material in the centers of galaxies. Independently of the precise physical mechanism of inflow, mass accretion to the central region builds up both the galaxy stellar mass and the SMBH mass. AGN feedback is then commonly invoked to quench star formation in a galaxy, shape the galaxy luminosity function and regulate the galaxy's

mass growth. However, the observations are still unclear about whether this combination of central inflows, and AGN feedback regulating SMBH and bulge growth, is indeed the driver behind galaxy - SMBH coevolution.

We propose to build on the science of JWST ERS program 1364 by doing a detailed study of the kinematics of the molecular and ionized gas in the nuclear region of NGC 4151 to test feeding and feedback processes in AGN. The science goals of this proposal are complimentary to those of ERS program 1364, which are focused on comparing black hole masses derived from the modeling of nuclear stellar dynamics to masses derived from reverberation mapping.

As one of the nearest AGN and an extremely well-studied Seyfert galaxy, NGC 4151 provides an outstanding opportunity to obtain meaningful constraints for models incorporating these processes to regulate SMBH-host galaxy coevolution. We will achieve this by: (1) tracing outflows via ionized and coronal gas emission to establish the energetics of the outflowing gas and potential for feedback into the host galaxy, (2) tracing inflows via molecular hydrogen gas emission from which the driving inflow mechanism can be identified and an inflow rate obtained via modeling, (3) using the wide range of available H₂ lines in the near-IR to establish their connection with the obscuring torus, and (4) obtaining a molecular gas-based black hole mass estimate. JWST is uniquely capable of advancing our understanding of these processes due to its sensitivity and stable point-spread-function, which facilitates the detection of diagnostic emission lines not available in current ground based data, as well as probing these processes down to unprecedented spatial scales.

NIRSpec combines great sensitivity with good spatial and spectral resolution and excellent, stable imaging quality (Strehl ratio > 0.8 at $2\ \mu\text{m}$), making it far superior than any 8-m class telescope working with AO. Using NIRSpec, we will be able to make use of IFU spectroscopy covering $1.6\text{--}3.2\ \mu\text{m}$ at a spectral resolution of ~ 2700 and angular resolution of $\sim 0.1''$.

Funding is requested for the areas of effort described above. The research activities outlined here will be carried out by the team members and one

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graduate student as soon as the JWST data becomes available (in communication with the ERS team).